A SEPARATING TRANSCENDENCE BASE EXISTS AMONG THE GENERATORS OF THE EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Minimal polynomial of a generator over the rational function field. Used at the start of the counted proof (lines 10528-10532). Produces the irreducible relation whose coefficients drive the argument.
- Separability tested by non-containment in the p-th power subring. Used at the end of the counted proof (lines 10564-10566). Converts the non-containment of the polynomial into separability of the extension.

Source reference. Original lemma 030F: sources/stacks-chow/tags.txt, source lines 3946–3946. Navigation pointer for \ref{fields-lemma-transcendence-degree} used inside the counted span (line 10555); the argument that consumes it is retained in the span.

Lemma 0.1. *Let k be a field of characteristic $p > 1$. Let K/k be a field extension generated by $x_1, \ldots, x_{n+1} \in K$ such that*

(1) *$\{x_1, \ldots, x_n\}$ is a transcendence base of K/k,*
(2) *for every k-linearly independent subset $\{a_1, \ldots, a_m\}$ of K the set $\{a_1^p, \ldots, a_m^p\}$ is k-linearly independent.*

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Then there is $1 \leq j \leq n+1$ *such that* $\{x_1, \ldots, \widehat{x}_j, \ldots, x_{n+1}\}$ *is a separating transcendence base for* K/k.

Proof. By assumption x_{n+1} is algebraic over $k(x_1, \ldots, x_n)$ so there exists a non-zero polynomial $F \in k[X_1, \ldots, X_{n+1}]$ such that $F(x_1, \ldots, x_{n+1}) = 0$. Choose F of minimal total degree. Then F is irreducible, because at least one irreducible factor must also have the same property.

We claim that, for some i, not all powers of X_i appearing in F are multiples of p. Suppose for a contradiction that all the exponents appearing in F were multiples of p, then the set

$$\{x_1^{\alpha_1} \ldots x_{n+1}^{\alpha_{n+1}} \mid \lambda_\alpha \neq 0\} \subset K$$

is k-linearly dependent where λ_α are the coefficients of F. By assumption (2) we conclude the set

$$\{x_1^{\alpha_1/p} \ldots x_{n+1}^{\alpha_{n+1}/p} \mid \lambda_\alpha \neq 0\}$$

is also k-linearly dependent, contradicting minimality of $\deg(F)$.

Choose i for which a non-pth power of X_i appears in F. Then we see that x_i is algebraic over $L = k(x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_{n+1})$. By Fields, Lemma 030F we see that $x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_{n+1}$ is a transcendence base of K/k. Thus L is the fraction field of the polynomial ring over k in $x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_{n+1}$. By Gauss' Lemma we conclude that

$$P(T) = F(x_1, \ldots, x_{i-1}, T, x_{i+1}, \ldots, x_{n+1}) \in L[T]$$

is irreducible. By construction $P(T)$ is not contained in $L[T^p]$. Hence K/L is separable as required. $\square$